MIL-T-36153A
9 February 1977
SUPERSEDING
MIL-T-36153 (BuMed)
28 May 1963

MILITARY SPECIFICATION

TEST TUBE, HEMATOCRIT DETERMINATION, WINTROBE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a disposable Wintrobe test tube suitable for use in military medical treatment facilities.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-541 - Glass (Laboratory).
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-packaging, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Defense Personnel Support Center by using the self-addressed Standardizatior Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Glass shall be in accordance with DD-G-541, Type I, Class A or B, or Type II.

3.2 Construction. Tube shall be transparent and shall have a uniform bore. Inside surface of the bottom shall be flat, and outside surface of bottom shall be round.

3.2.1 Dimensions. Dimensions shall be as shown in Table I.

TABLE I. Dimensions in millimeters (mm.)

Outside diameter	5.00 ± .25
Inside diameter	3.40 ± .50
Length	113.5 ± 3.5

3.2.2 Scale. Tube shall be marked with a scale which is at least 10 centimeters long. Scale shall start at the flat inner bottom and shall be marked with at least 100 evenly spaced lines. Every tenth line shall be numbered consecutively with Arabic numbers; numbers shall go in two directions. One set of numbers shall start 10 lines above the bottom and shall go upward from 1 to 10. The other set of numbers shall start 100 lines above the bottom and shall go down from 0 to 9. Numbered lines shall be longer than other lines and numbering of bottom line is optional. Additional lines may be marked above the scale. These lines shall conform to the characteristics of the graduation lines on the scale. Tolerance on overall length of scale shall be ± 1.5 mm. Overall length of scale shall be computed by adding theoretical distance for additional lines to 100 mm.

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3.2.2.1 Graduation marking. Graduation lines shall be of uniform width (maximum of .26 mm.). Graduation lines shall be perpendicular to the vertical axis of the tube and shall be parallel to each other. Graduation marking shall be green or white, shall be functionally legible and shall be sufficiently durable to withstand normal procedures (see 6.1). There shall be no change in marking when tested in accordance with 4.4.1.

3.3 Resistance to centrifugal force. When tested as specified in 4.4.2, test tube shall not chip, crack or break.

3.4 Workmanship. Workmanship shall be first class throughout. Tubes shall be free of defects listed in Table IV or any other defects which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determinatior by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the glass, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table II and MIL-STD-105. Unit of product for examination shall be one Wintrobe tube.

TABLE II. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table III and MIL-STD-105. Unit of product for test purposes shall be one tube. Acceptance number shall be zero.

TABLE III. For tests.

Characteristic	Requirement	Test procedure	Inspection level
Permanency of marking	3.2.2.1	4.4.1	S-2
Resistance to centrifuging	3.3	4.4.2	S-2

4.3 Examination. The tube shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to the defects listed in Table IV.

TABLE IV. Classification of defects.

Category	Defect
Major	
101	Tube not free of sharp edges.
102	Tube not free of cracks and breaks.
103	Open end not free of chipping.
104	Tube not free of protrusions.
105	Graduation marking not functionally legible. (Minor smears, smudges, and breaks shall not be considered cause for rejection.)

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TABLE IV. Classification of defects (Cont'd.)

Category	Defect
Major	
106	Graduation lines not evenly spaced.
107	Tube not free of dirt, powdered glass and other foreign substance.
Minor	
201	Tube not free of scratches.
202	Tube not free of checks.
203	Tube not free of stones.
204	Tube not free of air lines and blisters.

4.3.1 Dimensional examination. The tube shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Permanency of marking. Immerse tube in water for 30 seconds. Using a soft paper tissue, wipe graduated portion of the wet tube lightly with 10 complete up and down strokes.

4.4.2 Resistance to centrifugal force. Tube shall be filled with water to the top graduation and placed in a centrifuge. Speed and dimensions of centrifuge head shall be such that the inside bottom of the Wintrobe tube is subjected to a relative centrifugal force of not less than 2500 gravities. Formula for calculating relative centrifugal force is as follows:

$$RCF = .00001118 \times r \times N^2$$

RCF = relative centrigual force (gravities)
r = rotating radius to bottom of tube (centimeters)
n = rotating speed (revolutions per minute).

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified (see 6.2).

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5.1.1 Unit of issue. One package containing 500 tubes, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Tubes shall be cleaned in accordance with MIL-P-116, method C-1.

5.1.2.2 Drying. Tubes shall be dried in accordance with MIL-P-116.

5.1.2.3 Unit packaging. Five hundred tubes shall be shrink-wrapped and packaged in a plastic bag, or packaged by other suitable means in such a manner to permit withdrawal of one tube at a time, leaving the remaining tubes secure and protected from dust or other contamination. The wrapped tubes shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Tubes shall be clean, dry and packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Six unit packages (3000 tubes) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Six unit packages (3000 tubes) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged tubes shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Tube is intended for one-time hematocrit and blood sedimentation testing, including centrifuging.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

6.4 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-982-5979	TEST TUBE, HEMATOCRIT DETERMINATION Wintrobe, Disposable, 500s

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6640-0998

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-36153A TEST TUBE, HEMATOCRIT DETERMINATION, Wintrobe, Disposable

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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